The Media Connectivity Association



OVERVIEW AND UPDATES · BROADCAST AND MULTICAST

Media delivery over 5G Broadcast

The 3GPP-based broadcast delivery system bridging OTT streaming and broadcasting

3GPP ETSI

NOVEMBER 2026 – www.5g-mag.com

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About the project

The documents the 5G Broadcast project rests on

ETSI TS 103 720

Technical Specification

5G Broadcast System for linear TV and radio: the functions, reference points and receiver categories.

3GPP TR 36.976

Technical Report

The overall description of LTE-based 5G broadcast.

3GPP TS 26.346

Technical Specification

MBMS protocols and codecs. The MBMS application programming interface and URL are in 3GPP TS 26.347.

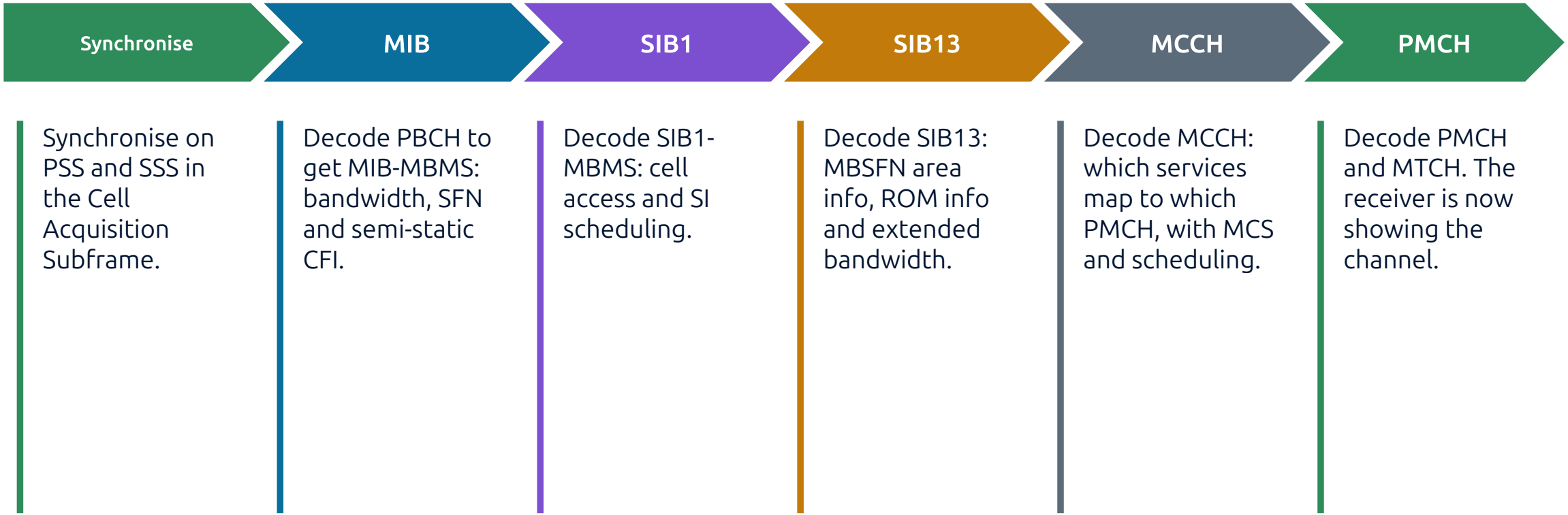
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How a receiver tunes in

From nothing to a channel on screen, in the order the receiver acquires the signal



There is no SIM, no return channel and no prior connection to the network: the receiver bootstraps itself.

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Features by release

What each 3GPP release added to LTE-based 5G Broadcast; ETSI TS 103 720 V1.2.1 profiles Releases 14 to 17

Release 14 the enTV baseline

- MBMS-dedicated cell
- 1.25 kHz numerology
- Cell Acquisition Subframe
- 256-QAM on PMCH
- MIB-MBMS and SIB1-MBMS
- Shorter MCCH periods

Release 16 LTE-based 5G Terrestrial Broadcast

- 2.5 kHz and 0.37 kHz numerologies
- PDCCH Format 4
- PBCH repetition
- MBSFN area info
- Semi-static CFI
- Receive-Only Mode

Releases 17 and 18 bandwidths (17), UHF bands (18)

- 6, 7 and 8 MHz PMCH bandwidths (17)
- Band 107 (18)
- Band 108 (18)

Release 19 PMCH Phase 2, CAS muting, new bands; not in V1.2.1

- Time interleaving
- Frequency interleaving
- PMCH cyclic shift
- Extended MCH scheduling periods
- CAS muting
- Bands 112 and 113

The UHF bands are receive-only (downlink-only). Release 20 (work plan of 28 September 2026): Band 246 (satellite) bandwidths, unverified.

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How the specification grew

5G-MAG has maintained ETSI TS 103 720 since 2021, with changes agreed among members

December 2020

First publication, with Release 16 as baseline: Free-to-Air and Receive-Only Mode services, single frequency networks over large distances, mobility up to 250 km/h, DASH, CMAF and HLS, IP-based services and file delivery.

June 2023

Release 17 alignment: receiver requirements for network planning, 6, 7 and 8 MHz bandwidths and broadcast UHF spectrum, 5G Media Streaming over eMBMS, public warning and emergency alerts based on cell broadcast.

In progress

The Release 18 update, started in 2024: DVB-I over 5G Broadcast, band support including Band 108, 5G Broadcast as a public warning improvement, concurrent 5G Broadcast and unicast, 3GPP Service URLs, a RAN-only receiver profile and low-latency distribution.

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Receiver categories

Four categories in ETSI TS 103 720; the fourth, the Emergency Alerts Receiver, has no Feature Value

Base receiver

Feature value 27

The minimum profile, aimed at fixed indoor reception. The 1.25 kHz numerology is mandatory; other subcarrier spacings are optional.

Main receiver

Feature value 28

Supports all five numerologies: 15, 7.5, 2.5, 1.25 and 0.37 kHz. The target for mobile and portable devices.

5GMS receiver

Feature value 30

Meets all the receiver requirements plus the 5G Media Streaming requirements, for 5GMS over LTE-based 5G Broadcast.

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The receiver chain

From the radio signal to a media player, as the Reference Tools receive it



The 5G Broadcast signal arrives as live I/Q data from an SDR, or from a prerecorded sample file.

Converts the signal into multicast IP packets on its output.

Decodes the FLUTE-delivered content from the multicast traffic and serves it to a media player.

Plays the stream in the browser with dash.js or hls.js.

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Repositories: transmitter and receiver

Transmitter, modem, middleware and web interface, from the descriptions of the repositories

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-mbms-tx	a standalone 5G Broadcast transmitter	Transmitter	● v1.0.0 21 Jun 2023	GitHub
rt-mbms-tx-for-qrd-and-crd	an MBMS-enabled eNodeB extended as a transmitter for QRD and CRD devices	Transmitter, QRD and CRD devices	● ea-1.0.0 5 Sep 2024	GitHub
rt-mbms-modem	converts an SDR or sample-file signal into multicast IP packets	Modem, SDR input, Sample files	● v1.2.1 30 Jun 2022	GitHub
rt-mbms-mw	decodes FLUTE content from the Modem and serves it from a web cache	Middleware, Web cache	● v0.10.0 8 Dec 2022	GitHub
rt-mbms-mw-android	an implementation of the MBMS Middleware for Android	Middleware, Android	● 1.0.0 5 Sep 2024	GitHub
rt-wui	a web interface that controls the Modem and Middleware and plays HLS	Web interface, HLS playback	● v0.1.0 8 Dec 2022	GitHub

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● released ○ no release yet

More information at 5g-mag.com/tech/5g-broadcast

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Repositories: core, applications and tools

The BM-SC, the Gateway, the Client, the applications and the tools, from the repository descriptions

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-mbms-bmsc	the BM-SC: ingests content, announces it and delivers it with FLUTE	BM-SC	none yet	GitHub
rt-mbms-gw	takes content from the BM-SC and delivers it onto the broadcast bearer	MBMS Gateway	none yet	GitHub
rt-mbms-client	reads the announcement, reassembles FLUTE objects, republishes HLS or DASH	MBMS Client	none yet	GitHub
rt-mbms-application	a web app that plays the MBMS Client output and shows radio state	MBMS-Aware Application	none yet	GitHub
rt-mbms-application-provider	a provider portal that provisions services and sessions on the BM-SC	Application Provider	none yet	GitHub
rt-libflute	FLUTE version 1, which carries files over the broadcast bearer	FLUTE	0.12.4 1 Sep 2026	GitHub
rt-mbms-examples	examples that use other 5G-MAG repositories, such as FLUTE ffmpeg	Examples, FLUTE ffmpeg	1.1.0 5 Nov 2024	GitHub

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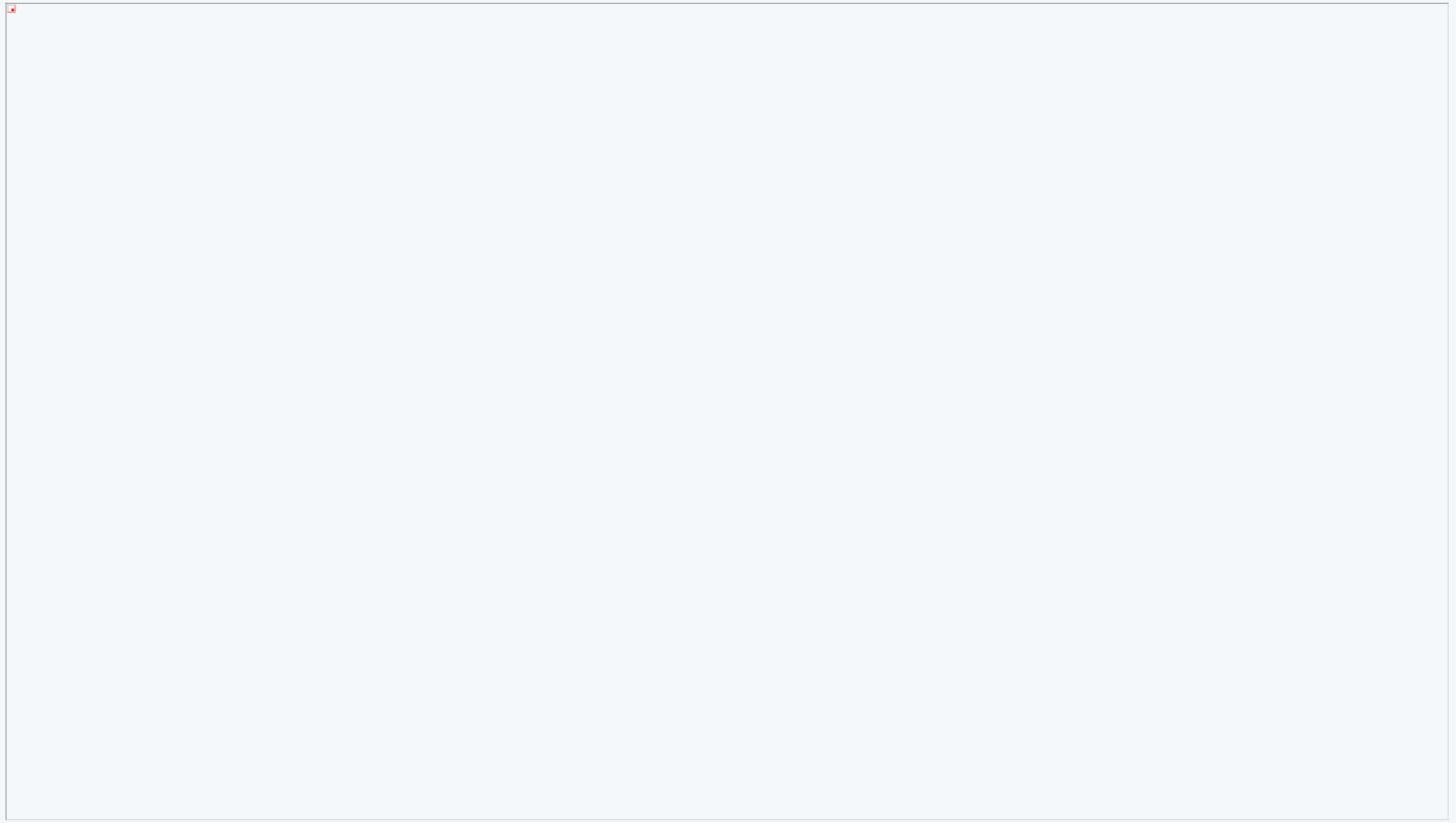
● released ○ no release yet

More information at 5g-mag.com/tech/5g-broadcast

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1. **rt-mbms-application**
2. **rt-mbms-application-provider**
3. **rt-mbms-bmsc**
4. **rt-mbms-client**
   used by 1
5. **rt-mbms-modem**
   used by 1
6. **rt-mbms-tx**
   used by 3, 7
7. **rt-mbms-gw**
8. **rt-libflute**
   dependency, used by 3, 4
9. **rt-common-shared**
   auxiliary library, used by 4

Boxes without a badge are not mapped to a repository. The MME has no repository.

See it and try it

A test setup with an SDR and phones, and the middleware app playing a live stream



SDR and phones, test setup



Middleware app

From the Reference Tools tutorials: seamless switching between broadcast and unicast on Android, and SDR playback.

WHERE WE STAND

Media delivery over 5G Broadcast

BROUGHT TO STUDY

- Integration of DVB-I with 5G Broadcast

ON-GOING

- Release 19 transmitter and receiver
- Docker set-up
- Validator for the BM-SC, the RAN and the signal

OUTCOMES

- LTE-based 5G Broadcast transmitter
- Seamless switching between 5G Broadcast and unicast
- Demonstration set-up for IBC 2024

RELATION WITH SDOS AND INDUSTRY FORA

3GPP ETSI

3GPP Potential proposals on satellite backhauling under discussion

ETSI TS 103 720 (5G Broadcast): maintenance, no progress in this period

More information at **5g-mag.com/standards/5g-broadcast**

5G MAG The Media Connectivity Association

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